Similarity Mapping and Factorized Vertical Hyperdiffusion in Atmospheric Boundary-Layer Modeling

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Abstract

This manuscript derives a local-equilibrium relation between gradient Richardson number and Monin–Obukhov stability, and uses Faà di Bruno's formula to map derivatives of a composite profile between similarity space and physical height. It also formulates a factorized fourth-order vertical filter whose continuous quadratic-norm dissipation follows for nonnegative coefficients and compatible boundary conditions. A smooth stability projection, roughness-sublayer taper, and weighted Tikhonov smoother are discussed as candidate methods requiring calibration. Their discrete conservation, boundary compatibility, effects on stable-boundary-layer profiles, and computational performance remain to be established for specified implementations; no forecast-validation or GPU-speedup results are claimed here.

1 Introduction, Mathematical Foundations, and HPC Operational Motivation

The atmospheric boundary layer (ABL) controls exchanges of momentum, heat, moisture, and trace constituents between the surface and free atmosphere. Stable-boundary-layer simulations remain sensitive to turbulence closures, surface coupling, vertical resolution, and numerical discretization [4, 20]. This study develops analytical relations for Richardson-number closures, coordinate derivative mappings, and a candidate factorized vertical filter. The proposed filter and regularization methods require implementation-specific verification and comparison with observations or established benchmarks before physical or computational benefits can be assessed.

1.1 The Operational Bottleneck in Boundary Layer Schemes

Atmospheric models distinguish numerical filtering from physical turbulent transport. Their horizontal and vertical operators vary by model, dynamical core, and configuration; no single formulation should be assumed across WRF, MPAS, FV3, and IFS. Many boundary-layer schemes use local or nonlocal closures for vertical turbulent fluxes [5, 10], whereas numerical filters address discretization errors and must be evaluated separately.

In stable conditions, some closures predict small eddy diffusivities, and some implementations impose lower bounds or other safeguards. Such choices are scheme-dependent and are not universally required to make a tridiagonal solve well posed. A positive lower bound can affect mixing and inversion structure; its impact depends on the closure, grid, surface forcing, and other transport processes. We therefore treat the effect of changing a diffusivity floor as a testable modeling question rather than a universal failure mode:

1. A floor may add mixing in weakly turbulent layers.
2. Its effects on inversion strength, jet structure, and surface coupling should be quantified against observations or benchmark simulations [7].
3. Removing a floor may also alter numerical conditioning or coupling; that effect must be checked for the particular discretization.

1.2 Computational and Kinematic Failure Modes

Beyond physical fidelity errors, standard boundary layer formulations induce severe computational bottlenecks and mathematical numerical instabilities on modern graphics processing units (GPUs) and vector accelerators:

1.2.1 HPC Warp and Thread Divergence

Some closures use piecewise functions or threshold logic. Divergent branches can reduce SIMD/SIMT efficiency when lanes take different paths, but branch coherence, compiler transformations, register use, and memory access also affect runtime. A smooth alternative may be worth benchmarking; smoothness alone does not establish faster execution or improved physical accuracy [8, 9].

1.2.2 The Zero-Shear Singularity

Near Low-Level Jet (LLJ) cores or elevated shear-free zones, horizontal wind shear vanishes ($S^2 \to 0$). The standard gradient Richardson number diagnostic, defined as:

$$Ri = \frac{\frac{g}{\Theta_v}\frac{\partial \Theta_v}{\partial z}}{S^2} \tag{1}$$

is undefined if its shear denominator vanishes while stratification remains nonzero, and becomes poorly conditioned when the shear is small. Error amplification

depends on wind and temperature uncertainties and on the derivative-estimation method; it is not characterized by a universal power law or a fixed floating-point threshold. Any regularization should state its scale, limiting behavior, and effect on the closure.

1.2.3 Geometric Confounding in Terrain-Following Grids

On stretched or terrain-following grids, discrete derivatives require the appropriate grid metrics. Grid stretching can change truncation error and conditioning, but it does not imply that the MOST stability parameter $\zeta = z/L$ has a stationary point. We use η for a computational vertical coordinate and reserve ζ for stability. Bell-polynomial chain rules apply to a specified smooth composition, such as $Ri_g(z) = R(\zeta(z))$; they do not by themselves remove grid error or diagnose turbulence extinction.

1.3 Objectives and Core Paper Thesis

This manuscript develops analytical relations and candidate numerical methods for evaluation. It does not assume that removing a diffusivity floor is universally appropriate:

1. **Richardson-number inversion:** An algebraic relation between Ri_g and ζ under stated similarity and local-equilibrium assumptions.
2. **Smooth stability regularization:** A zero-offset smooth projection as a candidate alternative to selected piecewise operations; closure behavior and hardware performance require testing.
3. **Geometric derivative mapping:** Fa'a di Bruno's formula for a specified coordinate composition, with inverse mapping restricted to nonsingular, well-conditioned cases.
4. **Factorized vertical hyperdiffusion:** A fourth-order filter whose discrete conservation, norm dissipation, boundary compatibility, and profile impacts must be verified for the chosen grid and implementation.

2 Non-Iterative C^∞ Monin–Obukhov Inversion, Intrinsic Similarity Derivatives $R^{(n)}(\zeta)$, and Z0HR Mechanics

To eliminate operational runtime bottlenecks caused by numerical iteration loops, boundary-layer schemes must explicitly transform implicit Monin–Obukhov Similarity Theory (MOST) [1] flux-profile relationships into explicit functions of the local gradient Richardson number (Ri). Converting implicit similarity functions into closed-form, non-iterative expressions provides predictable execution pipelines and optimal register usage on massively parallel architectures.

2.1 Exact Dual Transformation Identity

Monin–Obukhov Similarity Theory postulates that within the atmospheric surface layer, non-dimensional vertical gradients of mean wind speed U and virtual potential temperature Θ_v depend uniquely on the dimensionless stability parameter $\zeta = z/L$:

$$\phi_m(\zeta) = \frac{\kappa z}{u_*}\frac{\partial U}{\partial z}, \qquad \phi_h(\zeta) = \frac{\kappa z}{\theta_*}\frac{\partial \Theta_v}{\partial z} \tag{2}$$

where $\kappa \approx 0.40$ is the von K'arm'an constant, $u_* = (\tau_0/\rho)^{1/2}$ is friction velocity, $\theta_* = \frac{-\overline{w'\theta'_{v0}}}{u_*}$ is the surface friction temperature scale, and L is the Obukhov length scale:

$$L = \frac{u_*^2\Theta_v}{\kappa g\theta_*} = \frac{u_*^3\Theta_v}{\kappa g(-\overline{w'\theta'_{v0}})} \tag{3}$$

In localized gradient closures, turbulent exchange coefficients for momentum (K_m) and heat (K_h) are parameterized using stability shape functions $S_m(Ri)$ and $S_h(Ri)$ scaled by mixing length $l_m = \kappa z$:

$$K_m = (\kappa z)^2 S \cdot S_m(Ri), \qquad K_h = (\kappa z)^2 S \cdot S_h(Ri) \tag{4}$$

where $S = \left|\frac{\partial \mathbf{V}}{\partial z}\right| = \sqrt{\left(\frac{\partial U}{\partial z}\right)^2 + \left(\frac{\partial V}{\partial z}\right)^2}$ is the magnitude of vertical wind shear.

Theorem 2.1 (Local-Equilibrium Richardson–Similarity Relation)**.** *Under the stated local-equilibrium flux-gradient closure and similarity definitions, the mapping from gradient Richardson number Ri_g to the Monin–Obukhov coordinate ζ is:*

$$\zeta(Ri) = Ri\,\frac{S_h(Ri)}{S_m^{3/2}(Ri)} \tag{5}$$

Proof. Under local equilibrium, surface momentum and heat fluxes balance production:

$$u_*^2 = K_m S = (\kappa z)^2 S^2 S_m(Ri) \implies u_* = \kappa z S\sqrt{S_m(Ri)} \tag{6}$$

$$\overline{w'\theta'_v} = -K_h\frac{\partial \Theta_v}{\partial z} = -(\kappa z)^2 S\left(\frac{\partial \Theta_v}{\partial z}\right) S_h(Ri) \tag{7}$$

Substituting these exact flux expressions directly into the definition of the Obukhov length scale L:

$$L = \frac{u_*^3\Theta_v}{\kappa g(-\overline{w'\theta'_v})} = \frac{\left[\kappa z S\sqrt{S_m(Ri)}\right]^3\Theta_v}{\kappa g\left[(\kappa z)^2 S\left(\frac{\partial \Theta_v}{\partial z}\right) S_h(Ri)\right]} \tag{8}$$

Expanding the numerator and denominator powers:

$$L = \frac{\kappa^3 z^3 S^3 S_m^{3/2}(Ri)\Theta_v}{\kappa^3 z^2 g S\left(\frac{\partial \Theta_v}{\partial z}\right) S_h(Ri)} = z\left[\frac{\Theta_v S^2}{g\frac{\partial \Theta_v}{\partial z}}\right]\frac{S_m^{3/2}(Ri)}{S_h(Ri)} \tag{9}$$

Recognizing that $Ri^{-1} = \frac{S^2}{\frac{g}{\Theta_v}\frac{\partial \Theta_v}{\partial z}}$, the relationship simplifies to:

$$L = \frac{z}{Ri}\,\frac{S_m^{3/2}(Ri)}{S_h(Ri)} \tag{10}$$

Inverting L to form $\zeta = z/L$ completes the proof:

$$\zeta(Ri) = \frac{z}{L} = Ri\,\frac{S_h(Ri)}{S_m^{3/2}(Ri)} \tag{11}$$

□

Corollary 2.2 (Dimensional Consistency and Exponent Uniqueness)**.** *The 3/2 exponent follows algebraically from the cubic dependence of the Obukhov length on friction velocity, together with the chosen definitions of S_m and S_h. Since the stability functions, Ri_g, and ζ are dimensionless, this is not a dimensional-homogeneity argument. The relation is conditional on the stated flux and gradient assumptions, not a universal closure identity.*

2.2 Regime-Specific Analytical Inversions & Asymptotics

2.2.1 Stable Boundary Layer (SBL, $Ri > 0$)

Under stable stratification ($\zeta > 0$), classic linear similarity functions are defined as $\phi_m(\zeta) = 1 + \beta_m\zeta$ and $\phi_h(\zeta) = \alpha_\theta + \beta_h\zeta$, where $\alpha_\theta = Pr_{t,0}$ represents the neutral turbulent Prandtl number [2]. Expressing Ri in terms of ζ:

$$Ri(\zeta) = \zeta\frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} = \zeta\frac{\alpha_\theta + \beta_h\zeta}{(1+\beta_m\zeta)^2} \tag{12}$$

Cross-multiplying yields the quadratic mapping equation:

$$(\beta_m^2 Ri - \beta_h)\zeta^2 + (2\beta_m Ri - \alpha_\theta)\zeta + Ri = 0 \tag{13}$$

Applying the quadratic formula yields the explicit root branches:

$$\zeta^{\pm}(Ri) = \frac{(\alpha_\theta - 2\beta_m Ri) \pm \sqrt{\alpha_\theta^2 + 4(\beta_h - \alpha_\theta\beta_m)Ri}}{2(\beta_m^2 Ri - \beta_h)} \tag{14}$$

Lemma 2.3 (Branch Selection and Neutral Normalization)**.** *The physical branch governing SBL inversion is strictly given by the negative root (ζ^-).*

Proof. Evaluating the positive branch ζ^+ at $Ri = 0$:

$$\zeta^+(0) = \frac{\alpha_\theta + \sqrt{\alpha_\theta^2}}{2(-\beta_h)} = \frac{2\alpha_\theta}{-2\beta_h} = -\frac{\alpha_\theta}{\beta_h} \neq 0 \tag{15}$$

This violates the fundamental neutral physical boundary condition $\zeta(0) = 0$.

Conversely, direct substitution of $Ri = 0$ into the negative root branch ζ^- gives:

$$\zeta^-(0) = \frac{(\alpha_\theta - 0) - \sqrt{\alpha_\theta^2}}{2(-\beta_h)} = \frac{\alpha_\theta - \alpha_\theta}{-2\beta_h} = \frac{0}{-2\beta_h} = 0 \tag{16}$$

Because $\beta_h > 0$, the denominator $-2\beta_h \neq 0$. Thus, $\zeta^-(0) = 0$ is a determinate evaluation that directly satisfies the neutral boundary condition. Evaluating the derivative $\frac{d\zeta^-}{dRi}\Big|_{Ri=0}$ via quotient rule yields $\frac{1}{\alpha_\theta}$, confirming physical neutral normalization. □

2.2.2 Unstable Boundary Layer (UBL, $Ri < 0$)

Under unstable stratification ($\zeta < 0$), empirical functions take power-law forms [3]: $\phi_m(\zeta) = (1 - b_m\zeta)^{-1/4}$ and $\phi_h(\zeta) = \alpha_\theta(1 - b_h\zeta)^{-1/2}$. Expressing Ri in terms of ζ:

$$Ri(\zeta) = \alpha_\theta\zeta\sqrt{\frac{1-b_m\zeta}{1-b_h\zeta}} \tag{17}$$

Squaring both sides yields the Master Cubic Equation for unstable inversion:

$$\alpha_\theta^2 b_m\zeta^3 - \alpha_\theta^2\zeta^2 - b_h Ri^2\zeta + Ri^2 = 0 \tag{18}$$

Theorem 2.4 (The Exact Dyer Collapse)**.** *When empirical unstable exponents match ($b_m = b_h = 16$) and neutral stratification is unit-Prandtl ($\alpha_\theta = 1$), the unstable Monin–Obukhov mapping collapses identically to physical Richardson space ($\zeta \equiv Ri$).*

Proof. Setting $b_m = b_h = b = 16$ and $\alpha_\theta = 1$:

$$Ri = \zeta\sqrt{\frac{1-b\zeta}{1-b\zeta}} = \zeta\cdot 1 = \zeta \quad \Longrightarrow \quad \zeta(Ri) \equiv Ri \tag{19}$$

□

2.2.3 Lagrange Series Reversion Derivation

To prevent floating-point loss of precision near neutral stability ($\zeta \to 0$), $Ri(\zeta)$ is expanded via Taylor series:

$$Ri(\zeta) = r_1\zeta + r_2\zeta^2 + r_3\zeta^3 + \mathcal{O}(\zeta^4) \tag{20}$$

where $r_1 = \alpha_\theta$, $r_2 = \beta_h - 2\alpha_\theta\beta_m$, and $r_3 = \beta_m(3\alpha_\theta\beta_m - 2\beta_h)$. Applying the Lagrange Inversion Formula yields $\zeta(Ri) = a_1 Ri + a_2 Ri^2 + a_3 Ri^3 + \mathcal{O}(Ri^4)$:

$$a_1 = \frac{1}{\alpha_\theta}, \quad a_2 = -\frac{\beta_h - 2\beta_m\alpha_\theta}{\alpha_\theta^3} \tag{21}$$

$$a_3 = \frac{2(\beta_h - 2\beta_m\alpha_\theta)^2 - \alpha_\theta(3\alpha_\theta\beta_m^2 - 2\beta_m\beta_h)}{\alpha_\theta^5} \tag{22}$$

2.3 Closed-Form n-th Order Intrinsic Similarity Derivatives $R^{(n)}(\zeta)$

When constructing higher-order geometric transformations, derivative indices must be disambiguated from physical scale subscripts. Index n strictly denotes derivative order ($n = 1, 2, 3, 4$), whereas subscript m strictly denotes momentum scale properties (ϕ_m, β_m, S_m).

In physical Richardson space, the general linear mapping function $R(\zeta)$ under stable stratification ($\phi_m(\zeta) = 1 + \beta_m\zeta, \phi_h(\zeta) = \alpha_\theta + \beta_h\zeta$) is expressed as:

$$R(\zeta) = \zeta\frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} = \frac{\alpha_\theta\zeta + \beta_h\zeta^2}{(1+\beta_m\zeta)^2} \tag{23}$$

Theorem 2.5 (General Linear Derivative Theorem)**.** *For all integers $n \geq 1$ and $\beta_m \neq 0$, the nth derivative is:*

$$R^{(n)}(\zeta) = \frac{(-1)^n n!\,\beta_m^{n-2}\left[((n-1)\beta_h - n\alpha_\theta\beta_m) - \beta_m(2\beta_h - \alpha_\theta\beta_m)\zeta\right]}{(1+\beta_m\zeta)^{n+2}} \tag{24}$$

Proof. Define coordinate transformation $u = 1 + \beta_m \zeta$, which implies $\zeta = \frac{u-1}{\beta_m}$ and differential operator $\frac{d}{d\zeta} = \beta_m \frac{d}{du}$. Expressing $R(\zeta)$ in terms of u:

$$R(u) = \frac{1}{\beta_m^2}\left[\beta_h + \frac{\alpha_\theta\beta_m - 2\beta_h}{u} + \frac{\beta_h - \alpha_\theta\beta_m}{u^2}\right] \tag{25}$$

Differentiating $R(u)$ n times with respect to u ($n \geq 1$):

$$\frac{d^n R}{du^n} = \frac{(-1)^n n!}{\beta_m^2 u^{n+2}}\left[(\alpha_\theta\beta_m - 2\beta_h)u + (n+1)(\beta_h - \alpha_\theta\beta_m)\right] \tag{26}$$

Applying the chain rule $R^{(n)}(\zeta) = \beta_m^n \frac{d^n R}{du^n}$ and substituting $u = 1 + \beta_m\zeta$ completes the proof. □

Corollary 2.6 (Prandtl-Matched Invariant Manifold)**.** *When empirical coefficients satisfy the Prandtl-matched condition $\alpha_\theta\beta_m = 2\beta_h$ (derived from observational bounds $\beta_m \approx 5.0, \beta_h \approx 2.5, \alpha_\theta \approx 1.0$), the linear ζ term in the numerator vanishes identically. The derivative $R^{(n)}(\zeta)$ collapses to a pure inverse power law:*

$$R^{(n)}(\zeta) = \frac{(-1)^{n+1}\frac{(n+1)!}{2}\alpha_\theta\beta_m^{n-1}}{(1+\beta_m\zeta)^{n+2}} \tag{27}$$

2.4 Zero-Offset Hyperbolic Regularization (Z0HR)

To eliminate conditional GPU branching (`if/else`) across critical stability thresholds, physical Ri is decomposed into C^∞ smooth positive (Ri^+) and negative (Ri^-) projections using Zero-Offset Hyperbolic Regularization (Z0HR):

$$Ri^\pm = \frac{1}{2}\left(Ri \pm \sqrt{Ri^2 + \epsilon^2} \mp \epsilon\right) \tag{28}$$

Theorem 2.7 (Z0HR Error Bounds and Asymptotics)**.** *For $\epsilon > 0$, the positive projection is C^∞ and satisfies $Ri^+(0) = 0$. It is a smooth approximation to* $\max(0, Ri)$*, but its large-positive-Ri bias is $-\epsilon/2 + \mathcal{O}(\epsilon^2/Ri)$; it is not uniformly second-order accurate in ϵ relative to the unregularized positive part.*

Proof. At $Ri = 0$, $Ri^+(0) = 0$. For $Ri \gg \epsilon$, expansion gives $Ri^+ = Ri - \epsilon/2 + \epsilon^2/(4Ri) + \mathcal{O}(\epsilon^4/Ri^3)$. Thus the dominant asymptotic bias is $-\epsilon/2$. The regularization scale should be selected from the closure's sensitivity and the numerical precision used; it is not determined by comparing ϵ directly with machine epsilon. □

For the illustrative clamp $S_m(Ri) = [\max(0, 1 - Ri^+/Ri_c)]^2$, the derivative of Ri^+ at neutrality is $(Ri^+)'(0) = 1/2$, so $S_m'(0) = -1/Ri_c$, not $-2/Ri_c$. The clamp is C^1 at its extinction point because the squared factor and its first derivative vanish there; higher smoothness at the join does not follow. It should not be identified with a calibrated momentum stability function without further derivation.

2.5 Roughness Sublayer (RSL) Extension and Dual-Roughness Coupling

2.5.1 Roughness Sublayer and Dual-Roughness Extension

Monin–Obukhov similarity functions are conventionally applied above the roughness sublayer (RSL). Over tall vegetation and urban canopies, dispersive fluxes and canopy wakes make the layer immediately above the canopy distinct from the inertial surface layer; its depth depends on canopy structure and flow, rather than on a universal fixed multiple of canopy height [14,15]. An RSL correction is therefore a possible extension of the present framework, not a consequence of the Richardson-number inversion.

One way to define a differentiable transition for future testing is to introduce $s = z/z_{\mathrm{RSL}}$ and the standard smooth step

$$\eta(s) = \begin{cases} 0, & s \leq 0, \\ \exp(-1/s), & s > 0, \end{cases} \qquad \chi(s) = \frac{\eta(s)}{\eta(s) + \eta(1-s)}. \tag{29}$$

Then $\chi = 0$ for $s \leq 0$, $\chi = 1$ for $s \geq 1$, and χ is C^∞ at both joins. Candidate factors $\widehat{\phi}_a(s) = 1 - a_a[1 - \chi(s)]$, for $a \in \{m, h\}$ and $0 \leq a_a < 1$, can parameterize a gradual return to the above-canopy profile. This is a mathematical taper, not an empirically established canopy closure; z_{RSL}, a_m, and a_h require independent observational or LES calibration. Any modified gradients must be substituted into the flux and Richardson-number definitions and checked for dimensional and neutral-limit consistency before defining effective stability functions.

Momentum and scalar roughness lengths need not be equal. The excess-resistance parameter is often written $kB^{-1} = \ln(z_{0m}/z_{0h})$; its value is surface-dependent and should be specified or diagnosed for the application [4,17]. In neutral bulk-transfer form, with $L_m = \ln(z_1/z_{0m})$ and $L_h = \ln(z_1/z_{0h})$,

$$C_D = \frac{\kappa^2}{L_m^2}, \qquad C_H = \frac{\kappa^2}{L_m L_h}. \tag{30}$$

Under non-neutral conditions, integrated momentum and heat stability corrections enter the respective logarithmic profile differences. Their exact form depends on the chosen similarity functions and reference height; they should be derived from those profiles rather than appended as an unverified local $Ri_{1/2}$ factor. Canopy displacement height and the validity of the reference level above the canopy also need to be specified.

3 Geometric Derivative Mapping via Partial Bell Polynomials

In modern NWP architectures using terrain-following non-orthogonal vertical grid systems, non-uniform grid stretching causes spatial derivative operators to conflate coordinate geometry with intrinsic atmospheric thermodynamic curvature. To maintain scale-selective filtering without degrading boundary layer structures,

physical profile curvature must be disentangled from mesh distortion.

3.1 Fa'a di Bruno's Higher-Order Chain Rule in Stretched Boundary Layers

Consider the composite profile mapping $Ri(z) = R(\zeta(z))$, where $\zeta(z)$ represents the vertical grid coordinate profile. Spatial derivatives of Ri with respect to physical height z are governed by Fa'a di Bruno's higher-order chain rule expressed via partial exponential Bell polynomials $B_{m,n}$ [12, 13]:

$$\frac{d^m Ri}{dz^m} = \sum_{n=1}^{m} R^{(n)}(\zeta) \cdot B_{m,n}\left(g_1, g_2, \ldots, g_{m-n+1}\right) \quad (31)$$

where $g_j = \frac{d^j \zeta}{dz^j}$ represents the j-th spatial derivative of the vertical coordinate transformation, and $R^{(n)}(\zeta) = \frac{d^n R}{d\zeta^n}$ are the intrinsic similarity derivatives derived in Section 2.

3.2 Construction and Analytical Inversion of Partial Bell Matrix (B)

Up to fourth order, the vertical spatial derivative mapping is represented by the lower-triangular system $\mathbf{Ri}_z = \mathbf{B}\,\mathbf{R}_\zeta$:

$$\begin{bmatrix} Ri_z \\ Ri_{zz} \\ Ri_{zzz} \\ Ri_{zzzz} \end{bmatrix} = \begin{bmatrix} g_1 & 0 & 0 & 0 \\ g_2 & g_1^2 & 0 & 0 \\ g_3 & 3g_1g_2 & g_1^3 & 0 \\ g_4 & 4g_1g_3 + 3g_2^2 & 6g_1^2g_2 & g_1^4 \end{bmatrix} \begin{bmatrix} R'(\zeta) \\ R''(\zeta) \\ R'''(\zeta) \\ R^{(4)}(\zeta) \end{bmatrix} \quad (32)$$

Theorem 3.1 (Bell Matrix Determinant and Invertibility)**.** *The determinant of the 4th-order Partial Bell Matrix* $\mathbf{B}$ *is given by* $\det(\mathbf{B}) = g_1^{10}$.

Proof. Because $\mathbf{B}$ is strictly lower-triangular, its determinant equals the product of its diagonal elements: $\det(\mathbf{B}) = (g_1) \cdot (g_1^2) \cdot (g_1^3) \cdot (g_1^4) = g_1^{1+2+3+4} = g_1^{10}$. □

When $g_1 = \zeta_z \neq 0$, the analytical inverse matrix is:

$$\mathbf{B}^{-1} = \begin{bmatrix} g_1^{-1} & 0 & 0 & 0 \\ -g_2 g_1^{-3} & g_1^{-2} & 0 & 0 \\ (3g_2^2 - g_1 g_3) g_1^{-5} & -3g_2 g_1^{-4} & g_1^{-3} & 0 \\ (10 g_1 g_2 g_3 - 15 g_2^3 - g_1^2 g_4) g_1^{-7} & (15 g_2^2 - 4 g_1 g_3) g_1^{-6} & -6 g_2 g_1^{-5} & g_1^{-4} \end{bmatrix} \quad (33)$$

Applying $\mathbf{B}^{-1}$ recovers intrinsic derivatives only when the coordinate derivatives are known and the matrix is sufficiently well conditioned. Since $\det(\mathbf{B}) = g_1^{10}$, this inverse is singular at a fold ($g_1 = 0$) and can be severely ill-conditioned nearby; observations alone do not supply the required coordinate derivatives.

3.3 Kinematics of the Coordinate Fold ("The Fold Illusion")

A stationary point of the specified mapping $\zeta(z)$ occurs where $g_1 = \zeta_z = 0$. If $g_2 = \zeta_{zz} \neq 0$, the second derivative of the composed profile is $Ri_{zz} = R'(\zeta) g_2$ there; this does not imply that observed curvature is generally an artifact, nor that a terrain-following grid creates such a stationary point.

Theorem 3.2 (The Fold Illusion Theorem)**.** *At a stationary point with* $g_1 = 0$*, the* R'' *contribution to* Ri_{zz} *vanishes, while* $B_{2p,p}|_{g_1=0} = (2p-1)!!\, g_2^p$*. This is a chain-rule identity, not a physical attribution theorem; other Bell terms can remain and the inverse mapping is singular at the fold.*

3.4 Three-Layer Profile Curvature Decomposition

To prevent coordinate stretching artifacts from triggering unphysical vertical diffusion, raw model spatial curvature $\mathcal{M}_{\Delta z}[Ri_{zz}]$ is decomposed into a three-layer analytical model:

$$\mathcal{M}_{\Delta z}[Ri_{zz}] = \underbrace{R''(\zeta)\,\zeta_z^2}_{C_{\text{sim}}} + \underbrace{R'(\zeta)\,\zeta_{zz}}_{C_{\text{map}}} + \underbrace{\mathcal{E}_{\Delta z}}_{\text{discretization residual}} \quad (34)$$

The first two terms separate derivatives of the outer profile from derivatives of the mapping in this one-dimensional composition. Their signs are not fixed, and the residual depends on the discrete derivative operator. This identity alone does not establish that either term is a physical or numerical artifact.

3.5 Tikhonov–Morozov Regularization for High-Order Observational Inversions

3.5.1 Variance Gain Penalty (G) on Irregular Observational Arrays

Extracting intrinsic similarity derivatives $R^{(n)}(\zeta)$ from raw model output or field tower soundings requires inverting the lower-triangular Partial Bell Transformation Matrix $\mathbf{R}_\zeta = \mathbf{B}^{-1}\mathbf{Ri}_z$. Evaluating 3rd and 4th spatial derivatives of $Ri(z)$ using unconstrained finite-difference stencils on non-uniform vertical arrays ($\Delta z_i = z_{i+1} - z_i$) introduces severe observational error propagation.

For a linear stencil $D^{(m)} = \sum_i w_i y_i$ and independent measurement errors with variances σ_i^2, the derivative-estimate variance is:

$$\mathrm{Var}(D^{(m)} y) = \sum_{i=1}^{N} w_i^2 \sigma_i^2. \quad (35)$$

For homoscedastic errors, this reduces to $\sigma_{\text{obs}}^2 G$, where $G = \sum_i w_i^2$. Stencil bias and variance both depend on the stencil, grid, and measurement uncertainties; no universal variance increase follows solely from nonuniform spacing.

3.5.2 Nondimensional Tikhonov Operator Formulation

As a candidate smoother for tower profiles, define $\tilde{z} = z/H_{\text{SBL}}$, observed Richardson numbers $\mathbf{y}$, and a nonuniform-grid second-derivative matrix $\tilde{\mathbf{D}}_2$. For

known independent error variances, a weighted Tikhonov estimate minimizes [18]

$$\left\|\mathbf{W}^{1/2}(\mathbf{x}-\mathbf{y})\right\|_2^2+\tilde{\lambda}\left\|\tilde{\mathbf{D}}_2\mathbf{x}\right\|_2^2, \qquad \mathbf{W}=\mathrm{diag}(\sigma_1^{-2},\ldots,\sigma_N^{-2}), \quad \tilde{\lambda}\geq 0. \tag{36}$$

For fixed $\tilde{\lambda}$, the normal equations are $(\mathbf{W}+\tilde{\lambda}\tilde{\mathbf{D}}_2^{\mathsf{T}}\tilde{\mathbf{D}}_2)\widehat{\mathbf{x}}=\mathbf{W}\mathbf{y}$. The scaling of $\tilde{\mathbf{D}}_2$ and boundary constraints must be stated; with dimensionless height and a consistent discretization, $\tilde{\lambda}$ is dimensionless.

3.5.3 Dynamic Penalty Parameter Selection via Morozov's Discrepancy Principle

Morozov's discrepancy principle selects $\tilde{\lambda}$ so that the weighted residual is commensurate with the known noise level [19]. For independent Gaussian errors, the unregularized standardized noise norm has expectation N, so one possible target is $\tau^2 N$ with a stated tolerance factor $\tau \geq 1$:

$$\left\|\mathbf{W}^{1/2}(\widehat{\mathbf{x}}_{\tilde{\lambda}}-\mathbf{y})\right\|_2^2 \approx \tau^2 N. \tag{37}$$

This scalar equation need not have a solution for every data set and smoother. The selected parameter does not guarantee that real inversions are preserved; assess residuals, sensitivity, and resolution against independent information.

3.5.4 Continuous Sub-Grid Zero-Crossing Knee Detection

At a fold, $\mathbf{B}^{-1}$ is singular, so it cannot recover similarity derivatives there. If a coordinate stationary point is sought, estimate $\zeta(z)$ under an explicit closure assumption, fit a smooth local profile, and report the root uncertainty and sensitivity to the smoother. Linear interpolation between two samples is an estimate, not an exact sub-grid height; a curvature sign change is not by itself a coordinate fold or a physical transition.

4 Factorized Self-Adjoint Vertical Hyperdiffusion

As a numerical filter distinct from turbulent transport, fourth-order vertical hyperdiffusion can be formulated to damp short vertical wavelengths. Whether it permits reducing a scheme's background diffusivity floor or preserves resolved inversions must be tested for the coupled model.

4.1 Operator Formulation and Global Norm Dissipation

The 4th-order vertical hyperdiffusion operator is formulated in factorized self-adjoint form:

$$\mathcal{D}_4(q)=-\frac{\partial^2}{\partial z^2}\left[\mathcal{K}_z(z)\frac{\partial^2 q}{\partial z^2}\right]=-A^*\mathcal{K}_z A q \quad \text{where } A=\frac{\partial^2}{\partial z^2} \tag{38}$$

where $\mathcal{K}_z(z)\geq 0$ is the hyperdiffusivity coefficient (m^4/s) and A^* is the adjoint operator of A.

Theorem 4.1 (Global Variance Dissipation)**.** *For $\mathcal{K}_z(z)\geq 0$, a compatible operator domain gives nonincreasing quadratic norm $E_q=\frac{1}{2}\langle q,q\rangle$:*

Proof. Let $A=\partial_z^2$ with boundary conditions chosen so that the adjoint identity below holds, and define $\mathcal{D}_4=-A^*\mathcal{K}_z A$. Then

$$\frac{dE_q}{dt}=\langle q,\mathcal{D}_4 q\rangle=-\langle Aq,\mathcal{K}_z Aq\rangle=-\int_\Omega \mathcal{K}_z(z)\left(\frac{\partial^2 q}{\partial z^2}\right)^2 dz\leq 0. \tag{39}$$

This proves nonincrease, not strict decay: equality is possible when $Aq=0$ or the coefficient vanishes on the support of Aq. Boundary conditions are part of the operator definition and must also be compatible with the model's physical surface fluxes. □

4.2 Discrete Eigenvalue Recalibration

When projecting continuum hyperdiffusion operators onto discrete 3-point grid stencils, finite-difference truncation errors systematically attenuate peak damping rates.

For a uniform grid with spacing h, the shortest represented wavelength has $k=\pi/h$. At this mode the continuum biharmonic decay rate is $\omega_{\mathrm{cont}}=\nu_4(\pi/h)^4$, while the square of the centered three-point Laplacian has decay rate $\omega_{\mathrm{disc}}=\nu_4(4/h^2)^2$. Thus, with the same coefficient,

$$\frac{\omega_{\mathrm{disc}}}{\omega_{\mathrm{cont}}}=\frac{16}{\pi^4}\approx 0.1642. \tag{40}$$

The discrete operator therefore under-damps this mode by this ratio when its coefficient is copied unchanged from the continuum expression. Matching a specified continuum decay rate requires $\nu_{4,\mathrm{disc}}=(\pi^4/16)\nu_{4,\mathrm{cont}}$. This is a mode-specific calibration for this stencil, not a universal correction or evidence about profile accuracy.

4.3 Non-Uniform Staggered Grid Factorization & Boundary Clamping

On non-uniform vertical grids with cell thicknesses Δz_k and interface spacings $\Delta z_{k+1/2}=z_{k+1}-z_k$, 4th-order hyperdiffusion is executed via two sequential cell-centered curvature finite-difference steps:

1. **Cell-Centered Curvature Stencil (C_k):**

$$C_k=\frac{1}{\Delta z_k}\left[\frac{q_{k+1}-q_k}{\Delta z_{k+1/2}}-\frac{q_k-q_{k-1}}{\Delta z_{k-1/2}}\right] \tag{41}$$

2. **Face-Centered Hyperdiffusive Flux ($F_{4,k+1/2}$):**

$$F_{4,k+1/2}=\mathcal{K}_{k+1/2}\left[\frac{C_{k+1}-C_k}{\Delta z_{k+1/2}}\right] \tag{42}$$

3. **Net Cell Hyperdiffusive Tendency ($\mathcal{D}_{4,k}$):**

$$\mathcal{D}_{4,k}=-\frac{F_{4,k+1/2}-F_{4,k-1/2}}{\Delta z_k} \tag{43}$$

The boundary treatment must be specified as part of the discrete operator. Ad hoc curvature clamping does not, by itself, prove conservation, norm dissipation, or preservation of imposed surface momentum and heat fluxes. A useful construction is $\mathbf{D}_4 = -\mathbf{M}^{-1}\mathbf{A}^\mathsf{T}\mathbf{W}_K\mathbf{A}$, where $\mathbf{M}$ and $\mathbf{W}_K$ are positive mass and coefficient weights and $\mathbf{A}$ is the discrete curvature operator with boundary conditions included. Verify $\mathbf{A1} = 0$ for weighted-mean conservation and check the resulting tendencies and physical boundary fluxes separately.

4.4 CFL Stability Bounds & Numerical Solvers

For explicit Forward Euler time integration, the 4th-order vertical hyperdiffusion CFL stability bound is:

$$\Delta t \leq \frac{h^4}{8\nu_4} \tag{44}$$

Table 1: Comparison of Vertical Hyperdiffusion Numerical Solvers.

oprule extbf-Solver architecture	**Cost**	**Implementation considerations**
Pentadiagonal LU implicit	$\mathcal{O}(N)$ per solve	Requires a linear solve; performance depends on solver and hardware.
Super-time-stepping	Depends on stage count	May extend explicit stability intervals; accuracy and implementation costs must be measured.

5 Separating Closure, Coordinate, and Filter Diagnostics

Several distinct quantities appear in the analysis and should be reported separately. Their definitions do not imply that they are causally independent in a coupled model.

5.1 The Four Operational Spaces

The following categories organize diagnostics used in this study:

1. **Closure diagnostics:** Richardson number, stability functions, and closure parameters such as a specified Ri_c.
2. **Coordinate diagnostics:** Mapping derivatives g_j and the conditioning of the Bell transformation.
3. **Filter diagnostics:** The discrete operator spectrum, reference grid wavenumber, and damping time.
4. **Dynamical diagnostics:** Stability properties of a specified model system, if a governing state-space model and Jacobian are defined.

5.2 A Candidate Multi-Diagnostic Resolution Metric

As a proposed resolution diagnostic, track a height z_K at which eddy diffusivity falls below a stated threshold, a height z_e at which TKE falls below a stated threshold, and a height z_{e_z} of maximum TKE-gradient magnitude:

1. z_K: a threshold crossing defined using a documented K_m reference scale.
2. z_e: a threshold crossing defined using a documented TKE reference scale.
3. z_{e_z}: the height of maximum $|\partial e/\partial z|$ over a specified vertical interval.

Spatial spread dispersion C_{TP} is defined as:

$$C_{\mathrm{TP}} = \frac{\Delta z_{\mathrm{TP}}}{H_{\mathrm{SBL}}} = \frac{\max(z_K, z_e, z_{e_z}) - \min(z_K, z_e, z_{e_z})}{H_{\mathrm{SBL}}} \tag{45}$$

Report C_{TP} alongside the individual heights, threshold choices, and grid spacing. Its convergence may indicate that the selected diagnostics co-locate under refinement, but does not by itself verify physical turbulence extinction or distinguish model error from observational uncertainty.

6 Idealized Numerical Experiments & Validation

The following experiments are proposed to evaluate the framework. No model output or hardware benchmark is reported in this draft; quantitative conclusions should be added only after the configurations, data, and analysis are reproducible.

6.1 Benchmark Case Study: GABLS1 Nocturnal Boundary Layer

The GEWEX Atmospheric Boundary Layer Study stable-boundary-layer intercomparison provides a relevant single-column benchmark [7]. Compare a documented baseline closure with the proposed filter over a resolution sequence, stating the surface forcing, closure parameters, vertical grid, time step, boundary conditions, and any diffusivity lower bound. Assess wind and temperature profiles, surface fluxes, boundary-layer depth, and turbulent kinetic energy against the benchmark references. The effect of setting $K_{\min} = 0$ is a hypothesis to test, not a guaranteed outcome.

6.2 Suppression of $2\Delta z$ Stair-Casing

For each resolution, compare potential-temperature and wind profiles with and without the filter, and report inversion strength, jet height and speed, surface fluxes, and any grid-scale oscillation metric. A hyperdiffusive tendency is not a turbulence closure; evaluate it separately from the modeled turbulent fluxes and document whether its boundary treatment alters the prognostic state near the surface.

6.3 Computational Performance Benchmarking

To assess computational performance, benchmark equivalent kernels with the same input profiles, output fields, precision, compiler settings, and numerical tolerances. Report hardware and software versions, warm-up and timing procedures, workload size, memory transfers, and variability across repeated runs. Count divergent branches only as a hardware-counter metric where available; a branch-free expression does not imply zero divergence in the complete kernel or a speedup. No throughput or speedup is claimed here.

7 Operational Integration & Code Availability

7.1 Integration Hooks for Numerical Weather Prediction Models

The unified framework is designed for direct integration into existing operational weather prediction and Earth system modeling systems:

- **WRF-ARW:** A future integration could evaluate a smooth stability-function alternative within a selected boundary-layer scheme; compatibility and impacts would require scheme-specific testing.
- **MPAS-Atmosphere and FV3/GFS:** Porting would require adapting the operator to each model's grid, physics interface, and boundary treatment, followed by independent verification and validation.
- **Prototype implementations:** Small Julia or CUDA kernels can support isolated operator and performance tests. They do not establish integration into an operational model.

7.2 Reproducibility and Code Availability Statement

The manuscript's analytical derivations and any accompanying prototype code should be versioned with tests and documented build instructions before a reproducibility statement is made. At present, no operational model integration or benchmark repository is identified here.

- Archive the source, compiler or interpreter versions, input profiles, and test cases used to generate each reported result.
- For GPU measurements, include the exact device, software stack, kernel configuration, precision, and timing procedure.

7.3 Concluding Remarks

This manuscript develops analytical mappings and candidate numerical methods for atmospheric boundary-layer applications. The derivations identify useful relationships, while the proposed RSL extension, observational regularization, and vertical filter require implementation-specific testing. The present draft makes no claim of validated forecast improvement or measured computational acceleration:

1. Evaluate closure behavior, surface-flux compatibility, and stable-boundary-layer profiles against documented reference cases.
2. Verify discrete conservation, norm dissipation, boundary treatment, and resolution dependence for the chosen grid and time integrator.
3. Measure performance on specified hardware against an equivalent baseline before making claims about branch divergence or speedup.

A Inverse Derivative Mapping $Z^{(n)}(Ri)$ and Inversion Mechanics

The inverse mapping $\zeta = Z(Ri)$ forms an inverse Bell-matrix system governed by the inverse Fa'a di Bruno formula. The entries of the inverse partial Bell matrix $\mathbf{B}^{-1}$ derived for vertical coordinate stretching in Section 3.2 are dual to the higher-order derivatives of $Z(Ri)$ with respect to Ri.

A.1 Structural Link to the Inverse Bell Matrix

Setting $Ri = R(\zeta)$ and $\zeta = Z(Ri)$, repeated application of the chain rule yields $Z^{(n)}(Ri) = \frac{d^n\zeta}{dRi^n}$ in terms of the forward derivatives $R^{(k)} = \frac{d^kR}{d\zeta^k}$, evaluated at $\zeta = Z(Ri)$:

$$Z' = \frac{1}{R'}, \qquad Z'' = -\frac{R''}{R'^3}, \qquad Z''' = \frac{3R''^2 - R'R'''}{R'^5} \tag{46}$$

$$Z^{(4)} = \frac{10R'R''R''' - 15R''^3 - R'^2R^{(4)}}{R'^7} \tag{47}$$

With $g_k \equiv R^{(k)}$, the numerators and denominator power scalings $\left[g_1^{-1}, -g_2g_1^{-3}, (3g_2^2 - g_1g_3)g_1^{-5}, \dots\right]^T$ correspond directly to the first column of $\mathbf{B}^{-1}$ from Section 3.2.

A.2 General n-th Derivative Theorem

For $n \geq 2$:

$$Z^{(n)}(Ri) = \frac{1}{R'^n}\sum_{k=1}^{n-1}(-1)^k\frac{(n+k-1)!}{(n-1)!}\frac{1}{R'^k}, B_{n-1,k}\left(\frac{R''}{2}, \frac{R'''}{3}, \dots, \frac{R}{n}\right. \tag{48}$$

where $B_{n-1,k}$ represents the partial exponential Bell polynomial.

A.3 Closed-Form Derivatives

A.3.1 Unit-Prandtl Stable Regime

For $Pr_t = 1$, $\beta_m = \beta_h = \beta$, $\alpha_\theta = 1$, and $Ri_c = 1/\beta$, the inverse map reduces to a linear fractional transformation

$Z(Ri) = \frac{Ri}{1-\beta Ri}$. For all $n \geq 1$:

$$Z^{(n)}(Ri) = \frac{n!, \beta^{n-1}}{(1-\beta Ri)^{n+1}} = \frac{n!, Ri_c^{1-n}}{(1-Ri/Ri_c)^{n+1}} \quad (49)$$

A.3.2 Unstable Regime (Dyer Collapse)

For $Ri < 0$ under Dyer collapse conditions ($\phi_m = (1-16\zeta)^{-1/4}$, $\phi_h = (1-16\zeta)^{-1/2}$, $\alpha_\theta = 1$), the mapping simplifies to $Ri \equiv \zeta$. Thus $Z(Ri) \equiv Ri$:

$$Z'(Ri) = 1, \qquad Z^{(n)}(Ri) = 0 \quad \forall n \geq 2 \quad (50)$$

A.3.3 General Stable Branch

For $\gamma \equiv \beta_h - \alpha_\theta \beta_m \neq 0$ with profile functions $\phi_m = 1 + \beta_m \zeta$ and $\phi_h = \alpha_\theta + \beta_h \zeta$, the root satisfying $\zeta \to 0$ as $Ri \to 0$ is:

$$Z(Ri) = \frac{(\alpha_\theta - 2\beta_m Ri) - \sqrt{U(Ri)}}{2(\beta_m^2 Ri - \beta_h)}, \qquad U(Ri) = \alpha_\theta^2 + 4\gamma Ri \quad (51)$$

Higher-order derivatives follow from Leibniz's product rule applied to $N(Ri)/P(Ri)$, where $N(Ri) = (\alpha_\theta - 2\beta_m Ri) - \sqrt{U(Ri)}$ and $P(Ri) = 2(\beta_m^2 Ri - \beta_h)$. The k-th derivative of the radical term obeys:

$$\frac{d^k}{dRi^k}\sqrt{U(Ri)} = (-1)^{k-1}, 2^k, (2k-3)!!, \gamma^k, U(Ri)^{-\frac{2k-1}{2}} \quad (52)$$

with the double-factorial convention $(-1)!! \equiv 1$.

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